

Using n8n to Build an AI-Based Habit Tracking System

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Abstract—This paper presents the design and implementation of an AI-powered habit tracking system using n8n as a workflow automation platform. The system integrates form-based data collection, Google Sheets storage, scheduled data processing, and generative AI analysis using Google Gemini. The objective is to automate habit monitoring and provide personalized behavioral feedback based on user activity data. The proposed system demonstrates how low-code automation tools such as n8n can be combined with AI services to build intelligent productivity applications efficiently. Experimental workflow design shows that the system successfully collects user input, processes structured datasets, and generates meaningful behavioral recommendations automatically.

Keywords—AI automation, n8n, habit tracking, workflow automation, Google Sheets, Gemini AI, low-code systems

I. INTRODUCTION

Habit tracking systems have become increasingly important in productivity optimization and behavioral analytics. Traditional systems require manual input and lack intelligent feedback mechanisms. With the rise of workflow automation tools and large language models, it is now possible to build systems that not only store data but also analyze and interpret user behavior.

This paper proposes an AI-based habit tracking system built using n8n, an open-source workflow automation tool that allows integration of APIs and services without complex backend development [1].

II.I. SYSTEM DESIGN AND ARCHITECTURE

Overview

The proposed system architecture is designed based on the principle of modular workflow automation, where each process is separated into independent but interconnected components. This design choice is made to ensure scalability, maintainability, and ease of integration with external services such as Google Sheets and AI APIs.

The overall system is divided into two primary workflows. This separation is intentionally introduced to distinguish between data acquisition phase and data analysis phase, allowing the system to first focus on reliable data collection before performing computationally intensive AI-based

processing. As a result, the system achieves a clear cause-and-effect pipeline: user input is systematically transformed into structured datasets, which are then interpreted by an AI model to generate meaningful behavioral insights. The system consists of two main workflows.

Workflow 1: Data Input System

As attached in Fig. 1 the first workflow is responsible for the initial stage of data acquisition. It is implemented using an n8n Form Trigger node, which enables users to submit their daily activity information in a structured format. The decision to use a form-based input mechanism is based on the need for simplicity and consistency in user-generated data.

Once the data is submitted, the system immediately appends the input into a Google Sheets database. This approach is chosen due to its lightweight nature and real-time accessibility, allowing the dataset to serve as a centralized storage layer for subsequent processing stages.

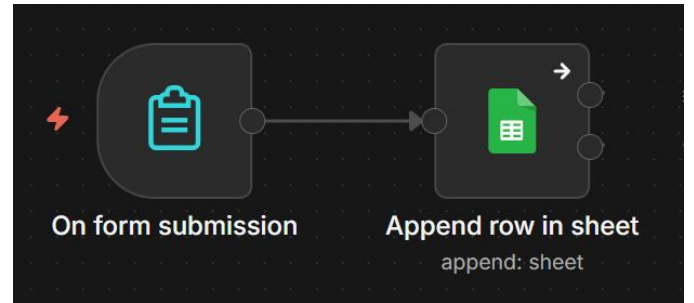


Fig. 1. Input workflow architecture of the AI habit tracking system implemented using n8n automation.

The first workflow can be summarized as:

- Uses an n8n Form Trigger node
- Collects:
 - Activity name
 - Duration (hours)
- Stores data into Google Sheets

Workflow 2: AI Analysis System (result)

As seen in Fig. 2 the second workflow operates as the analytical core of the system and is executed on a scheduled basis (07:00 daily). The use of a scheduled trigger is intended to ensure periodic evaluation of accumulated user data without requiring manual intervention.

At this stage, data is retrieved from Google Sheets and transformed using a JavaScript-based preprocessing step. This transformation is necessary to convert raw tabular data into a structured JSON format suitable for AI model consumption.

The processed dataset is then forwarded to the Google Gemini language model via the n8n AI Agent node. The selection of a large language model is motivated by its ability to perform contextual reasoning, pattern recognition, and natural language generation based on user behavioral data.

Finally, the generated analysis output is delivered to the user via Gmail automation. This completes the end-to-end pipeline, where raw user activity is progressively transformed into actionable insights through a structured workflow chain.

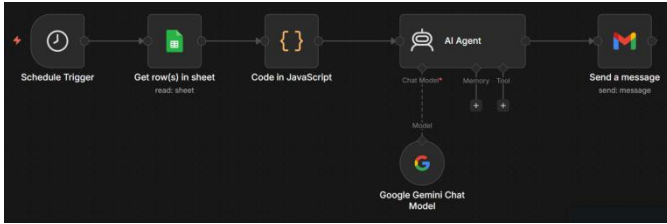


Fig. 2. End-to-end result workflow architecture of the AI habit tracking system implemented using n8n automation.

The second workflow can be summarized as:

- Runs on a scheduled trigger (07:00 daily)
- Fetches data from Google Sheets
- Processes data using JavaScript transformation
- Sends structured data to Google Gemini AI
- Sends output via Gmail

II.II. SYSTEM ARCHITECTURE

The design of this architecture is based on a sequential data transformation principle, where raw user inputs are progressively processed into structured and meaningful insights. This design choice is made to ensure that each stage of the system has a clear responsibility, thereby reducing complexity and improving system reliability.

The workflow begins with user input collected through an n8n Form Trigger, which ensures standardized and consistent data acquisition. The collected data is then stored in Google Sheets, which acts as a lightweight and accessible data repository.

Following data accumulation, a scheduled trigger activates the second workflow at a predefined time (07:00 daily). This

timing decision is intended to allow sufficient data collection before analysis is performed.

Once triggered, the system retrieves stored records and performs preprocessing using a JavaScript transformation node. This step is necessary to convert tabular spreadsheet data into a structured JSON format suitable for machine interpretation.

The processed dataset is then forwarded to the Google Gemini AI model for behavioral analysis. The model identifies activity patterns and generates personalized recommendations based on the aggregated data.

Finally, the generated insights are delivered to the user via automated email notification, completing the full automation loop.

As a result of this structured separation, the system follows a logical cause-and-effect pipeline: user-generated activity data is first captured through an input interface, then stored in a centralized database, and subsequently processed using scheduled automation. This processed data is finally analyzed using an AI model to generate personalized behavioral feedback.

The system architecture can be summarized as:

User Input → n8n Form → Google Sheets → Scheduled Trigger → Data Processing → AI Analysis (Gemini) → Email Output.

III.I. IMPLEMENTATION: DATA COLLECTION MODULE

The Form Trigger node in n8n enables real-time data entry. Each submission is appended to a Google Sheets document using the Google Sheets API integration [2].

III.II. DATA PROCESSING MODULE

A JavaScript function transforms raw spreadsheet rows into structured JSON format:

- activity name
- duration
- timestamp

III.III. AI ANALYSIS MODULE

The structured dataset is forwarded to the Google Gemini model (gemini-2.5-flash) through the n8n AI Agent node for further semantic processing and behavioral analysis. This design choice is based on the need to leverage large language models for interpreting unstructured human activity data into meaningful insights without requiring manually defined rules.

By utilizing the generative and reasoning capabilities of the model, the system is able to infer higher-level interpretations from raw activity logs. Specifically, the model generates:

- estimated calorie-related interpretation based on activity type and duration,
- time distribution analysis across different daily activities,
- and personalized health recommendations derived from behavioral patterns.

This transformation process demonstrates how large language models can be applied beyond traditional natural language processing tasks, extending their capability to behavioral pattern analysis and decision-support systems. Previous research has shown that transformer-based language models are capable of performing complex reasoning tasks through learned contextual representations [4].

As a result, raw user activity data is effectively converted into actionable insights, enabling automated self-monitoring and continuous habit improvement feedback.

III.IV. NOTIFICATION MODULE

The notification module serves as the final stage of the proposed system, where the analyzed results are delivered directly to the user through automated email services. This module is implemented using the Gmail node provided by n8n, which is connected through the Google Gmail API for secure and reliable communication [1], [5].

The integration with Google's API ecosystem plays an important role in ensuring interoperability between different services used within the system. Since the data storage component already utilizes Google Sheets, the use of Gmail API creates a unified cloud-based workflow environment that simplifies authentication, data exchange, and automation management [2]. This architectural arrangement is designed to minimize manual intervention while improving synchronization between data processing and notification delivery stages.

After the AI analysis process is completed by the Google Gemini model, the generated output is automatically formatted into a readable text-based report. The report contains summarized activity statistics, estimated calorie-related information, total time spent on each activity, and personalized behavioral recommendations generated by the AI model [3].

This result is then transmitted to the user via email without requiring manual interaction from the system administrator. The use of automated email delivery is intentionally selected to improve accessibility and user convenience, as users can receive their daily habit analysis directly through commonly used communication platforms. Previous studies have shown that automated feedback mechanisms can positively influence user engagement and self-monitoring behavior in digital habit-tracking systems [6].

In addition, the notification mechanism contributes to the practical usability of the system by enabling continuous monitoring and feedback cycles. As a result, users are encouraged to consistently evaluate their daily habits and make healthier behavioral adjustments based on the AI-generated recommendations.

From a system architecture perspective, the notification module also demonstrates how API-based service integration can be utilized to create a fully automated end-to-end intelligent application using low-code technologies such as n8n [1].

IV. RESULTS AND DISCUSSION

The system successfully automates habit tracking and provides AI-generated insights without manual analysis. The combination of n8n and Gemini AI reduces development complexity while increasing functionality. However, system accuracy depends on data quality and prompt engineering.

V. CONCLUSION

This study demonstrates that n8n can effectively serve as a backbone for AI-powered automation systems. The proposed habit tracking system integrates data collection, processing, and AI-based analysis into a single workflow. Future improvements may include mobile integration and more advanced behavioral prediction models.

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REFERENCES

- [1] n8n GmbH, "n8n Workflow Automation Platform," <https://n8n.io>
- [2] Google, "Google Sheets API Documentation," <https://developers.google.com/sheets/api>
- [3] Google, "Gemini API Documentation," <https://ai.google.dev/gemini-api>
- [4] T. Brown et al., "Language Models are Few-Shot Learners," NeurIPS, 2020.
- [5] Google, "Gmail API Documentation," <https://developers.google.com/gmail/api>
- [6] B. J. Fogg, *Persuasive Technology: Using Computers to Change What We Think and Do*, Morgan Kaufmann, 2003.